

Vascular contributions to cognitive impairment and dementia (VCID): A report from the 2018 National Heart, Lung, and Blood Institute and National Institute of Neurological Disorders and Stroke Workshop

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[This article has been contributed to by US Government employees and their work is in the public domain in the USA.]

Abstract

Vascular contributions to cognitive impairment and dementia (VCID) are characterized by the aging neurovascular unit being confronted with and failing to cope with biological insults due to systemic and cerebral vascular disease, proteinopathy including Alzheimer's biology, metabolic disease, or immune response, resulting in cognitive

decline. This report summarizes the discussion and recommendations from a working group convened by the National Heart, Lung, and Blood Institute and the National Institute of Neurological Disorders and Stroke to evaluate the state of the field in VCID research, identify research priorities, and foster collaborations. As discussed in this report, advances in understanding the biological mechanisms of VCID across the wide spectrum of pathologies, chronic systemic comorbidities, and other risk factors may lead to potential prevention and new treatment strategies to decrease the burden of dementia. Better understanding of the social determinants of health that affect risks for both vascular disease and VCID could provide insight into strategies to reduce racial and ethnic disparities in VCID.

KEYWORDS

Alzheimer's disease-related dementias, cerebrovascular and cardiovascular disease, vascular contributions to cognitive impairment and dementia

1 | INTRODUCTION

The National Alzheimer's Project Act, signed into law in January 2011, calls for an aggressive and coordinated national plan to accelerate research on Alzheimer's disease (AD) and Alzheimer's disease-related dementias (ADRD), and to provide better clinical care and services for people living with dementia and their families. The first national plan was created in 2012. The plan establishes five ambitious goals to both prevent future cases of AD/ADRD and to better meet the needs of the millions of American families currently facing dementia. The National Institutes of Health (NIH) leads the research effort on the first goal, to prevent and effectively treat AD/ADRD by 2025. To help achieve this goal, triennial AD, ADRD, and dementia care summits are held to assess and refine research priorities in the national plan. Vascular contributions to cognitive impairment and dementia (VCID), a term used to describe the broad field of research considering the role of a range of vascular risk and vascular lesions on cognition and used broadly by the American Heart Association,¹ the Alzheimer's Association,² and the NIH,³ is defined as an ADRD, and thus research priorities for VCID are set at the ADRD summits. The last ADRD summit held prior to this workshop was the ADRD Summit 2016 (see Table 1, summary of the AD and ADRD summits). To evaluate the state of the field in VCID research, refine and identify new priorities for VCID, and foster collaborations, the National Heart, Lung, and Blood Institute (NHLBI) and the National Institute of Neurological Disorders and Stroke (NINDS) convened a working group (WG) May 7 to 8, 2018.⁴ The WG brought together a group of interdisciplinary experts and program staff from NIH to build upon recommendations from the VCID research milestones defined by ADRD Summit 2016.⁵ In this report, we provide a summary of the WG presentations, discussion, and its recommendations for future research. This report reflects the attendees' perspectives on current understanding on VCID research, potential mechanisms contributing to VCID, and research gaps as well as future research opportunities, and is not meant to be a comprehensive review.

We include a limitation section at the end of the report to acknowledge several important research areas that were not discussed in the WG or, as a result, this report. Finally, for the purposes of this report, we use the term VCID throughout to refer to "vascular contributions to cognitive impairment and dementia," as this has been the preferred use of this term in recent years. This is distinct from specific clinical syndromes, such as vascular cognitive impairment, and our use of the term VCID referring to contributions rather than purely vascular syndromes is felt to better represent the great deal of overlapping pathologies seen, with a range of clinical and radiographic manifestations. In addition, the term VCID allows inclusion of a broad range of vascular mechanisms and phenotypes, as is appropriate for this heterogeneous group of disorders. These range from monogenetic disorders leading to a pure small vessel disease phenotype to cerebral amyloid angiopathy to mixed pathologies, with many forms between. This report is focused on common pathways and mechanisms that occur in VCID, as well as research priorities for this heterogeneous group of disorders.

Dementia is a constellation of neurological and vascular disorders that typically occur during aging and are characterized by cognitive impairment severe enough to interfere with one's daily function. While AD is the most common dementia diagnosis, AD as a clinical syndrome displays multiple pathologies in more than half of clinically diagnosed individuals. Emerging evidence suggests that most clinical AD cases, especially those in persons over the age of 80, have both classic AD pathology with plaques and tangles as well as vascular pathology.⁶ VCID is defined as the neurovascular unit being confronted with and failing to cope with biological insults, often as people age, due to systemic and cerebral vascular disease, proteinopathy including Alzheimer's biology, metabolic disease, and/or immune response, resulting in cognitive decline.² VCID research overlays multiple clinical diagnoses, including cerebro- and cardiovascular disease, stroke, AD and other types of dementia, some of which involve neither AD nor stroke.³ Chronic systemic comorbidities and risk factors are also associated with deleterious cognitive outcomes. These comorbidities and

risk factors include but are not limited to: cardio⁷ and cerebrovascular disease;⁸ metabolic disorders of lipid and glucose metabolism including diabetes and obesity;⁹⁻¹¹ immune dysfunction;^{12,13} and behavioral risk factors for many dementia comorbidities such as smoking, dietary behavior, physical inactivity, and possibly stress.¹⁴⁻¹⁶ The racial/ethnic and socioeconomic health disparities in VCID have been described yet are not well understood, and often are consistent with health disparity patterns in other age-related comorbidities. We expect that promoting a fuller and deeper understanding of the mechanisms of VCID at molecular, cellular, and systems levels across the wide spectrum of pathologies, chronic systemic comorbidities, and other risk factors may lead to potential prevention and treatment strategies to decrease the burden of dementia.

2 | CURRENT STATE OF KNOWLEDGE

The WG meeting included review of the existing state of knowledge in the field of VCID, with a focus on epidemiologic and basic science evidence supporting vascular contributions to dementias of aging. This report represents the knowledge reviewed and synthesized by the working group members, and thus is not a fully comprehensive review covering all aspects of VCID. Furthermore, the format of the WG did not incorporate public comment, and thus we acknowledge that this document is a reflection of the material covered rather than a representation of a broader interpretation of the existing literature.

2.1 | Epidemiology of VCID

There is a large and growing body of epidemiological evidence suggesting an association between cardio- and cerebrovascular risk factors and disease; and cognitive impairment and dementia diagnoses.¹⁷⁻¹⁹ While some risk factors may directly contribute to dementia, others may have a more indirect impact, possibly through affecting risks for cardio- and cerebrovascular diseases (either clinically apparent or quiescent). As the ability to diagnose "subclinical" cardio- or cerebrovascular diseases improves, the understanding of potential mechanisms linking vascular risk factors and markers and dementia expands. However, epidemiologic data clearly suggest that the potential impact of vascular risk factors on dementia may not be entirely through easily measurable vascular disease (such as stroke or small vessel disease or silent brain infarcts), or at least as limited by current technology. Microvascular dysfunction, measured using magnetic resonance imaging (MRI) as well as biomarkers of microvascular disease, only accounted for 16.2% of the effect of aortic stiffness on cognitive function,²⁰ white matter hyperintensities (WMH) only partially mediated the effect of hypertension on cognition,²¹ and mediation analyses of other vascular risk factors have not found a clear imaging feature explaining the relationship between risk factor and cognition.²² As technology and biomarkers of downstream vascular changes of these risk factors improves, more will be understood about the full range of mechanisms at play relating to these risk factors. In addition, it is also plausible that a vascular risk

RESEARCH IN CONTEXT

1. Systematic review: Recognizing the important contribution of vascular risk factors to dementia, this working group was convened to identify research priorities that complement and refine the vascular contributions to cognitive impairment and dementia (VCID) Research Milestones defined by the National Institutes of Health 2016 Alzheimer's disease-related dementias (ADRD) Summit and to guide VCID research.
2. Interpretation: The working group presentations and discussions were focused on three areas: (1) known vascular factors which may lead to dementia, (2) optimal methods to study VCID, and (3) potential mechanisms of VCID. The working group identified some important research gaps and opportunities which are included in the article.
3. Future directions: We hope that the report can promote research to better understand biological mechanisms of VCID across the wide spectrum of pathologies, chronic systemic comorbidities and other risk factors, with an ultimate goal of leading to potential prevention and treatment strategies to decrease the burden of dementia.

factor might act more directly on AD pathology, as suggested in the Atherosclerosis Risk in Communities-Positron Emission Tomography (ARIC-PET) study using PET imaging²³ and in the Honolulu-Asia Aging Study by autopsy studies,²⁴ while other studies have failed to show a similar association,²⁵ making it imperative to study the vascular risk factors themselves and not only their downstream end-organ injuries.

Given the frequency of comorbidities, particularly vascular risk factors, the association between any single risk factor and dementia is hard to discern in human observational studies. A primary advantage of data provided from observational studies, however, is the ability to evaluate the dynamic changes of risk factors and human physiology over the life course, which are more challenging to evaluate in clinical trials or even animal models, given the need to observe patterns or changes potentially over decades. Additionally, the combined effect of multiple risk factors for dementia seems to be more important and a combined risk score has been shown to be a stronger predictor of cognitive decline than independent risk factors;²⁶⁻²⁸ a composite vascular risk score was also associated with brain amyloid when individual risk factors were not, in the above-cited ARIC-PET study.²³ The CAIDE (Cardiovascular Risk Factors, Aging and Dementia study) risk score strongly predicted late-life dementia, when considered in midlife,²⁹ and had an area under the curve of 0.74 for dementia prediction in a study of Finnish twins.³⁰ Life course is especially relevant because risk factors appear to be most strongly associated with adverse cognitive outcomes when considered in middle age, or perhaps even earlier: mid-life exposure (45 to 64 years) to hypertension, smoking, and diabetes has a greater impact on dementia risk than similar risk factors in

TABLE 1 Links to research recommendations of the past AD/ADRD summits

AD Summit		ADRD Summit	
2012	https://www.nia.nih.gov/news/alzheimers-disease-research-summit-offers-research-recommendations	2013	https://aspe.hhs.gov/alzheimers-disease-related-dementias-adrd-conference-2013-prioritized-research-recommendation-milestones
2015	https://www.nia.nih.gov/research/agenda-alzheimers-disease-research-summit-2015	2016	https://aspe.hhs.gov/adrd-summit-2016-report-national-advisory-neurological-disorders-and-stroke-council
2018	https://www.nia.nih.gov/research/administration/recommendations-nih-ad-research-summit-2018	2019 ^a	https://www.ninds.nih.gov/sites/default/files/2019_adrd_summit_recommendations_508c.pdf

Abbreviations: AD, Alzheimer's disease; ADRD, Alzheimer's disease-related dementias.

^a2019 ADRD summit occurred after this workshop.

late-life (>65 years),¹⁷⁻³⁵ with a near-doubling of dementia risk from midlife diabetes (hazard ratio [HR] 1.77, 95% confidence interval [CI] 1.53 to 2.04), a > 40% increased risk of dementia for midlife smokers (HR 1.41, 95% CI 1.23 to 1.61), and a similarly elevated risk for individuals with hypertension in middle age (HR 1.39, 95% CI 1.22 to 1.59), in the ARIC study,¹⁷ with even higher estimates for more pronounced increases in blood pressure (HR 4.8, 95% CI 2.0 to 11.0 in the Honolulu Heart Program for midlife blood pressure 160 mm Hg and higher and risk of dementia).³⁶

Another important use of observational data is in evaluating trends in populations over time.¹⁷ The incidence of dementia in high-income countries has declined for the past decades, which has occurred in parallel to the improvement of cardiovascular health, and it has been postulated that better prevention and management of vascular diseases and risk factors might partially contribute to this decline.^{37,38} This does not, however, mean a reduction in dementia prevalence, given the estimated longer life span of individuals with dementia, and thus it has been hypothesized that there will be a doubling of individuals with ADRDs by 2060, compared to 2014.³⁹

Despite the concerning rising prevalence of dementia, the reduction in incidence may be informative and ultimately, if continued, would lead to reductions in prevalence. Thus, a better understanding of what factors contribute to the decline could provide insight into possible prevention and intervention strategies. The SPRINT-MIND (Systolic Blood Pressure Intervention Trial-Memory and Cognition in Decreased Hypertension) trial, completed after this workshop, showed that intensive blood pressure control lowered risk for mild cognitive impairment (MCI)—although this was not the study's primary outcome; no significant effect was found for dementia, the primary outcome—strengthening the connection between vascular health and brain health and providing momentum to examine whether targeting other vascular risk factors or multiple risk factors could prevent VCID.⁴⁰ As further evidence that risk factors such as hypertension do not act entirely through measurable brain lesions or brain disease, in the SPRINT-MIND MRI sub-study, white matter lesions (WML), interpreted as small vessel disease, progressed less rapidly in individuals treated with more aggressive blood pressure control with statistically significant but relatively small effect size. At the follow-up (median intervention period of 3.4 years and median follow-up time

of 3.97 years), the intensive treatment group had less increase in WML compared to the standard treatment group yet the difference between the two groups was small, with a mean between-group difference for change in WML volume of -0.54 cm^3 (95% CI, -0.87 to -0.20 cm^3).^{40,41} The FINGER (Finnish Geriatric Intervention Study to Prevent Cognitive Impairment and Disability) trial demonstrated that a multimodal intervention, which included both lifestyle interventions as well as vascular risk factor control, across multiple risk factors, was promising in reducing 2-year cognitive decline,⁴² further demonstrating the potential for prevention by focusing on vascular contributions. Further studies modeled after FINGER are under way across the world, and will provide important information on prevention of dementia through vascular risk.⁴³ As most of the vascular risk factors are modifiable, preventable, and treatable, it is attractive to target vascular contributors of dementia as potential prevention and new treatment options. It is possible, however, that not all preventive therapies should be applied equally to all individuals, and intensive blood pressure control may be ideal in most but not all adults, with remaining questions about ideal treatment targets in the oldest old or individuals with especially high cerebrovascular risk.

2.2 | VCID pathophysiology

As stated above, vascular pathology frequently accompanies AD pathology in clinical AD patients based on human autopsy findings, especially in patients of older age.⁴⁴ Vascular disease and vascular tissue injury, such as intracranial atherosclerosis, arteriolosclerosis, silent infarcts, microinfarcts, cerebral amyloid angiopathy, and microbleeds, have been shown to correlate with dementia symptoms, with less AD pathology observed for an equivalent level of cognitive impairment if vascular pathology is also present.⁶ However, the mechanisms by which these vascular pathologies may contribute to dementia are not well understood. Potential mechanisms linking vascular function to cognitive function and potential effects of vascular disorders on cognitive impairment have been summarized in recent reviews.^{3,45,46} Cerebrovascular abnormality is thought to occur early in the AD continuum, as depicted in the revised Jack model of AD biomarkers.^{45,47}

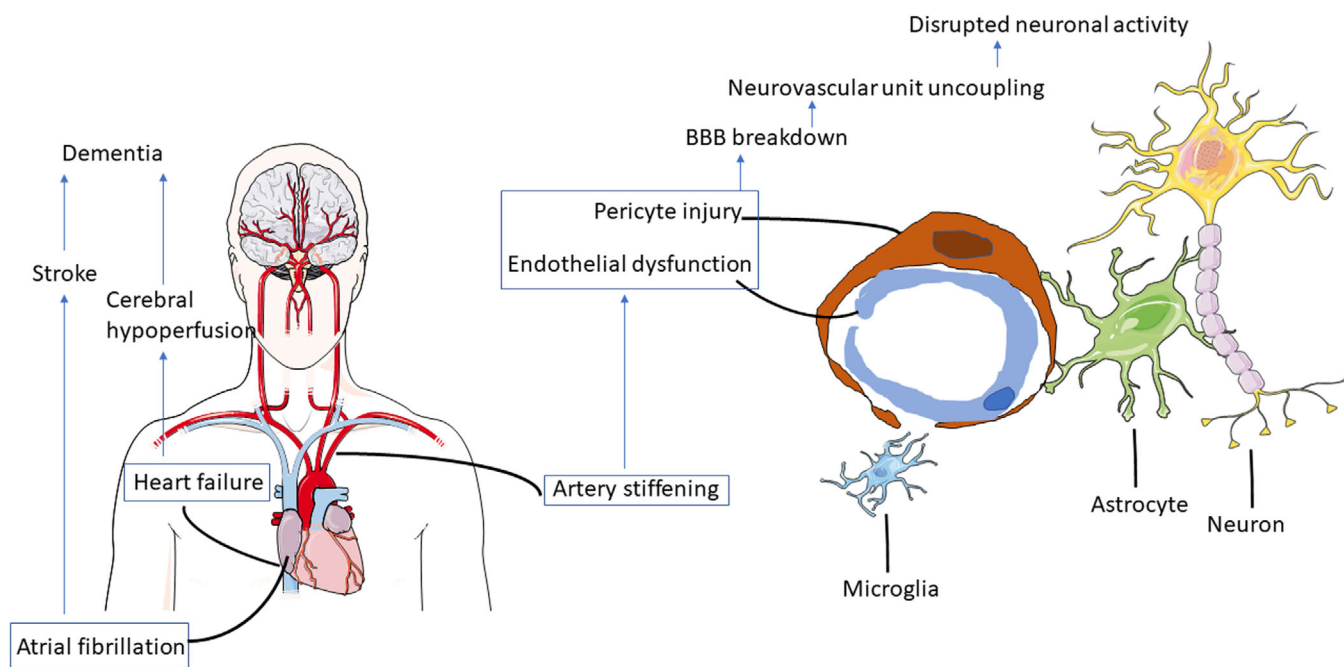


FIGURE 1 Linking peripheral vascular risk factors and cardiovascular diseases to cognitive impairment and dementia. This figure presents examples illustrating how pathology originating from the peripheral vasculature or the heart could lead to cerebral vascular abnormality or affect cerebral blood flow and disrupt neuronal activity and increase risk for dementia. Arterial stiffening as a result of loss of elasticity in large blood vessels could lead to cerebral vasculature damage, including pericyte injury and endothelial dysfunction, which in turn causes blood-brain barrier (BBB) breakdown and neurovascular unit uncoupling. Heart failure could cause chronic cerebral hypoperfusion, an important risk factor for dementia. Atrial fibrillation could increase risk for stroke, which is known to cause dementia

However, peripheral vascular deficits or other systemic metabolic changes may precede cerebrovascular abnormality in some cases. If we could identify mechanisms of cerebrovascular diseases and early changes in the peripheral vasculature, it may offer opportunities of finding early biomarkers for detecting cognitive decline (see Figure 1 for examples of impact of peripheral vascular risk factors or cardiovascular diseases on cognitive function).

Studies using animal models, imaging techniques, and *post mortem* studies in humans have revealed several potential mechanisms underlying VCID, including: hypoperfusion and hypoxia, increased blood-brain barrier (BBB) permeability, endothelial dysfunction, systemic inflammation and oxidative stress, and trophic uncoupling.^{45,46} The brain, just like the heart, has high metabolic demand and requires steady blood supply to maintain its structural and functional integrity. Cerebral hypoperfusion is thought to be one mechanism that can lead to cognitive decline.⁴⁸ A variety of systemic and local dysfunctions could affect cerebral perfusion, including diseases such as arrhythmias, ischemic heart disease and heart failure, chronic lung disease, and sleep apnea. Disruption of BBB caused by dysfunctional endothelium has been noted in patients with cognitive decline. Oxidative stress and inflammation induced by several vascular risk factors (hypertension, insulin resistance, and diabetes) contribute to endothelial dysfunction and disruption of trophic coupling.⁴⁶ In addition to contributing to cognitive impairment independently, vascular dysfunction could also amplify AD pathology through stimulating the production and limiting the clearance of misfolded amyloid beta (A β) and tau^{46,49,50} (see

Table 2 for selected mechanisms linking vascular function to neurovascular unit [NVU] function).

An important step in testing the VCID hypothesis is to develop biomarkers for early diagnosis. Typical dementia has a long pre-symptomatic phase that has been proposed as the optimal time window for intervention. Several imaging-based measures are candidate biomarkers for VCID, as they can, or have been proposed to, detect early vascular changes that occur before the appearance of clinical symptoms, including WMH, infarcts and microinfarcts, BBB disruption, resting cerebral blood flow (CBF), and increased cerebrovascular resistance.⁵¹ Recent studies have introduced and evaluated potential biomarkers for BBB leakage, which has been reported to be an early biomarker of human cognitive dysfunction independent of A β and tau.⁵² Novel cerebral spinal fluid (CSF) biomarkers for BBB breakdown, such as soluble platelet-derived growth factor receptor- β ,⁵⁰ and imaging methods such as dynamic contrast enhanced MRI, hold promise in measuring BBB permeability in humans.^{52,53} Additionally, a multimodal imaging study highlighted vascular dysfunction as a potential upstream trigger of late-onset AD in a multimodal imaging study,⁵⁴ highlighting the potential of using vascular function and dysfunction tests that could be measured outside the brain, such as a retinal vascular imaging exam.⁵⁵ It is important to validate these biomarkers to better understand the correlation between local and systemic clinical manifestations as well as the correlation between their utility and interpretation in animal models and humans. With the advancement of imaging probes and technologies, and a better understanding of the biological

TABLE 2 Potential mechanisms linking vascular function to neurovascular unit function

Mechanisms	Capillary function	Function within NVU	Effects of vascular disorders
Hemodynamic flow	Hemodynamic pressure gradients (osmotic and hydrostatic)	Nutrient delivery to organ/tissue	Ischemia, infarction, capillary rarefaction
Structural integrity	Tight junctions	BBB integrity	Solute leakage
Neurovascular coupling	Astrocyte function, pericyte function, vascular distensibility, etc.	Autonomic innervations and Functional hyperemia	Decoupling of vascular function and neuronal activity
Nutrient extraction point and trophic support	Diffusion, facilitate transport and maintain the vascular-neuron trophic coupling	Nutrient delivery (eg, O ₂ , glucose, ketone bodies) and sensing of CO ₂ and pH changes	Impaired metabolism and functional response
Drainage	Collection site	Facilitates solute clearance	Solute aggregation

Abbreviations: BBB, blood-brain barrier; NVU, neurovascular unit.

basis of VCID, more molecular and cellular biomarkers are expected to emerge to diagnose VCID or stratify patients. Another important aspect of new biomarker development is the need to understand how to best translate various biomarkers from experimental animal models to humans. The heterogeneity of vascular disorders means that no single animal model could recapitulate the full spectrum of VCID. Although there are animal models capturing one or two features of the disease, such as hypoperfusion injury, focal ischemia, hypertension, or hyperhomocysteinemia (HHcy) based on the main disease pathway featured by each of these animal models, more diverse experimental models are needed to capture mixed pathology, late onset, and progressive nature seen in the majority of dementia patients. While rodent models are useful, their limitations are also apparent. Because rodents have different brain structures, neurobiology, and immune systems than do humans, replicating results from rodents in a different experimental model would help increase the translational value of the findings from animal studies to human studies.⁵⁶ Alternative experimental models, such as tissue chips and induced pluripotent stem cells, could also be used to study BBB and interaction among various cell types. A multi-pronged approach to take advantage of various experimental models and innovative technologies would be needed to understand the complex pathophysiology and mechanisms of VCID.

3 | POTENTIAL MECHANISMS

3.1 | Endothelial dysfunction

The cerebrovascular endothelium plays critically important roles in regulation of CBF, BBB, blood coagulation, immune surveillance, and trophic support of neuronal and glial cells. Historically, the concept of "endothelial dysfunction" was articulated in the 1980s, shortly after discovery of endothelium derived relaxing factor/nitric oxide (NO).⁵⁷⁻⁶⁰ Loss of NO is considered to be a central mechanism of endothelial dysfunction.⁶¹ Indeed, major cardiovascular risk factors including hypertension, diabetes, hypercholesterolemia, and aging impair the production and/or biological activity of endothelial NO. Relevant to this report, existing literature supports the notion that

endothelial dysfunction may contribute to initiation and development of cognitive impairment through a variety of mechanisms. A study using cultured human brain microvascular endothelial cells suggested that inhibition of endothelial nitric oxide synthase (eNOS) causes significant upregulation of amyloid precursor protein (APP) as well as expression and activity of β -site amyloid precursor protein cleaving enzyme 1.⁶² Further analysis of signaling mechanisms demonstrated that the effects of NO on expression and processing of APP are mediated by activation of soluble guanylate cyclase and increased production of cyclic guanosine monophosphate (GMP).^{62,63} In vivo validation and extension of the results obtained in cultured human cerebrovascular endothelium was performed in eNOS-null mice (Nos3^{-/-}). Consistent with in vitro findings, elevated expression of APP, beta-secretase 1 (BACE1), and increased production of A β peptides were detected in the brain parenchyma of eNOS-null mice.^{62,64} Moreover, it was found that endothelial NO deficiency instigates activation of microglia and impairment of glycolytic function of astrocytes.^{65,68} Behavioral studies revealed that eNOS-null mice suffer from cognitive impairment.^{65,66} The importance of intact eNOS function in preservation of cognitive function was also demonstrated in an experimental model of endothelial dysfunction induced by a high salt diet.^{69,70} While the exact mechanisms responsible for cognitive impairment induced by the loss of endothelial NO are complex and incompletely understood, studies from several groups indicate that besides aberrant processing of APP, inactivation of eNOS (by gene deletion or treatment with high salt diet) also promotes phosphorylation of tau protein in neuronal cells.^{67,70,71} Importantly, in mice exposed to a high salt diet, NO precursor, L-arginine, suppressed phosphorylation of tau and prevented cognitive dysfunction. In contrast, NO deficiency reduced neuronal calpain nitrosylation that in turn caused activation of cyclin-dependent kinase 5 and tau phosphorylation.⁷¹ These observations suggested that reduction in CBF (hypoperfusion) does not explain detrimental effects of eNOS inactivation on phosphorylation of tau and cognitive function and that endothelium-derived NO may diffuse into brain parenchyma and exert direct effects on neuronal cells.⁷⁰ Additionally, partial eNOS-deficiency in aged (eNOS^{+/-}) mice first leads to elevation of A β 40 in cerebral blood vessels rather than in the brain parenchyma,⁷² suggesting that in this model, cerebrovascular alterations occur independently

of changes within the brain. Furthermore, it appears that endothelial dysfunction might also contribute to pathogenesis of cerebral amyloid angiopathy (CAA). Indeed, aged eNOS^{+/−} mice develop CAA and CAA-associated pathology including smooth muscle cell degeneration, microhemorrhages, and cognitive impairment.⁷¹ In aggregate, these observations provide molecular mechanisms that may help explain increased occurrence of dementia in patients exposed to cardiovascular risk factors.⁷³ In humans, genetic variants of eNOS and soluble guanylate cyclase, enzymes responsible for enhanced production and signaling of endothelial NO, are associated with reduced risks of stroke, coronary artery disease, and peripheral arterial disease.⁷⁴ These findings support the concept that therapeutic interventions designed to stimulate endothelial NO signaling may protect function of the cerebral, coronary, and systemic circulation. Whether preservation of vascular endothelial function in humans reduces risk of dementia is still unknown, and thus remains to be investigated.⁷⁵

3.2 | Pericyte dysfunction

Pericytes are contractile cells located between brain capillary endothelial cells (ECs), astrocytes, and neurons.⁷⁶ Their unique position suggests that pericytes may regulate key neurovascular functions of the brain, which has been supported by recent findings. Pericyte loss has been shown to reduce CBF and brain capillary perfusion, and diminish global and individual capillary responses to neuronal stimuli, which results in neurovascular uncoupling and reduced oxygen supply to the brain and metabolic stress.⁷⁷ Furthermore, pericyte loss also has been demonstrated to cause BBB breakdown and accumulation of blood-derived toxin proteins in the brain.⁷⁸ Rapid loss of pericytes could also lead to loss of pericyte-derived neurotrophic growth factors pleiotrophin (PTN) and acute circulatory collapse. Notably, it has been reported that intracerebroventricular PTN infusion prevented neuron loss in pericyte-ablated mice despite persistent circulatory changes; thus, suggesting that restoration of pericyte function might be a potential therapeutic option.⁷⁹ Not surprisingly, disrupted white-matter microcirculation occurs before neuronal loss in pericyte-deficient mice, suggesting that pericytes play an important role in regulating white-matter structure and function. A recent study showed that A β oligomers can constrict capillaries both in humans and mice specifically at pericyte locations, but not in arterioles and venules,⁸⁰ highlighting the crucial role pericytes play in NVU and pericytes as an important mediator of A β oligomer-induced vascular damage. Interestingly, BBB breakdown and elevated CSF levels of pericyte injury marker sPDGFR β (soluble platelet derived growth factor receptor- β) predicted cognitive decline in apolipoprotein E (APOE)4 (ϵ 3/ ϵ 4 or ϵ 4/ ϵ 4) carriers independently of AD A β and tau pathology in a recent study.⁸¹ Consistent with data from the mouse,⁸² this study found that APOE4 (ϵ 3/ ϵ 4 or ϵ 4/ ϵ 4) carriers have elevated CSF levels of proinflammatory cyclophilin A (CypA) and matrix metalloproteinase-9 (MMP9) suggesting activation of this BBB-degrading pathway by APOE4 gene, and that pericytes derived from APOE (ϵ 4/ ϵ 4) human induced pluripotent stem cells had higher levels of CypA and secreted MMP9 than those

derived from APOE (ϵ 3/ ϵ 3), implicating that pericyte function may be affected by APOE genotype.⁸¹ In addition to pericytes, there is evidence that brain perivascular macrophages also affect BBB permeability and other aspects of neurovascular regulation.^{83,84}

3.3 | Hypertension

Several lines of investigation are converging to suggest blood pressure (BP) control may prevent dementia and associated neuropathology.^{85,86} Indeed hypertension and diabetes are recognized as the strongest modifiable risk factors for AD/ADRD.⁸⁷ A growing volume of observational studies show associations between vascular risk factors (eg, hypertension and arterial stiffness) and evidence of AD pathology (eg, A β and tau).^{88,89} It is important to note that measured BP is a complex phenotype bolstered by autonomic factors, peripheral resistance, and structural changes in the vasculature. While associative findings do not prove causality, there is direct evidence in animals and humans indicating that better cardiovascular health (including lower BP) is advantageous for brain health.^{90–92} Results from the SPRINT MIND clinical trial demonstrated that intensive BP control (systolic BP target of ≤ 120 mm Hg vs 140 mm Hg) may reduce the risk of MCI, which is often a prodromal phase of dementia.⁴⁰ Finally, specific medications used to control hypertension have been hypothesized to have beneficial cognitive effects, beyond an effect of reducing cognitive decline purely by reducing blood pressure. This may be particularly true for drugs active on the renin-angiotensin system, such as angiotensin II receptor blockers and angiotensin converting enzyme (ACE) inhibitors.⁹³ These drugs may have additional effects beyond those acting on blood pressure, including anti-inflammatory effects and possible direct impacts on A β degradation.^{94,95} Recent meta-analyses, however, do not show a clear antihypertensive drug class effect for dementia prevention.^{96,97}

Recent experiments in mice have emphasized that immune modification by ACE can greatly affect the pathology observed in models of AD. Specifically, experiments with ACE 10/10 mice, a strain that overexpresses ACE in monocytes and macrophages, showed that such mice have a more effective immune response when challenged with either tumors or models of infection.⁹⁸ These mice were crossed with a mouse model of AD and compared to AD mice with wild type expression of ACE.⁹⁹ The animals overexpressing ACE in macrophages had a marked and highly significant reduction in cortical amyloid plaque area, reduced astrogliosis, reduced soluble A β levels, and retention of cognitive function. These data emphasize that ACE-mediated enhancement of the immune response may be effective at reducing AD-associated neuroinflammation and A β pathology. They also emphasize how important it is to understand the effects on AD of all drugs used to treat hypertension and other cardiovascular problems. Finally, these data suggest that it may be useful to consider inflammation as not just the number of inflammatory cells or even the types of cells comprising the infiltrate, but rather by some measure of effectiveness of the inflammation in clearing the underlying biological insult.

3.4 | Atherosclerosis

An examination of temporal trends in the age-specific incidence of dementia in the Framingham Heart Study and other cohorts showed that the age-specific incidence of dementia has declined over the past three decades.¹⁰⁰ The parallel improvement in several vascular risk factors (hypertension, hyperlipidemia, and smoking) raised the possibility that this decline might be attributable to reduced vascular risk factors; however, adjusting for vascular risk factors at midlife or after age 65 did not substantially alter the trend.¹⁰¹ Atherosclerotic cardiovascular events such as stroke and myocardial infarcts, however, did partially explain the temporal trend observed. This apparent contradiction might be explained by considering subclinical and clinical manifestations of atherosclerosis as markers of accumulated exposure to vascular risk factors over a lifetime, or by considering the impact of atherosclerosis on brain, cardiac, and other organ structure and function.¹⁰²

The extent of systemic atherosclerosis tends to be correlated across various arterial beds and can be readily quantified. Cerebral or systemic atherosclerosis has been related to MRI evidence of structural brain injury and poorer cognition in many cohort studies of community-living men and women, across various race and ethnic groups.¹⁰³⁻¹⁰⁶ These effects seem to begin as early as the third and fourth decades of life and the adverse impact of early atherosclerosis on brain structure and function seems even greater than the effect of atherosclerosis later in life,¹⁰⁷⁻¹¹¹ emphasizing that VCID accrue throughout life; that is, dementia in at least some instances is a life course disease.

Atherosclerosis may adversely impact brain structure and function through multiple pathophysiological mechanisms. The atheroma can directly alter the stiffness of the aorta and cerebral vasculature; arterial stiffness assessed as carotid-femoral pulse wave velocity has been associated with smaller brain volumes, increased volume of abnormal white matter, and microstructural injury within apparently normal looking white matter.^{107,112-114} It could also impair cerebral circulation at rest or under stress even at subclinical degrees of carotid stenosis.¹¹⁵ Coronary atherosclerosis and myocardial infarction, perhaps through ischemic cardiomyopathy, can lower cardiac index and hence cerebral perfusion^{116,117} and has also been associated with overt and covert cardioembolic brain infarctions,¹¹⁸ either directly or due to shared risk and atherosclerosis in extra- or intracranial vascular beds paralleling coronary disease.

Atherosclerosis may also be viewed as a paradigm that offers valuable lessons for understanding VCID. Like atherosclerosis, VCID may begin early in life. Adherence to the American Heart Association Life's Simple 7 optimal lifestyle characteristics reduces risks, not only of atherosclerotic cardiovascular disease events, but also of cognitive decline and dementia and like atherosclerosis many genetic and biological pathways, including inflammation, oxidative stress, and mitochondrial dysfunction, seem to underlie most dementia.^{119,120} In fact, common genetic pathology has been shown to underlie both cardiovascular disease and dementia and both conditions share many circulating biomarkers such as homocysteine, leptin, clusterin, YKL-40,

CD14, miRNA 132, BNP, ST2, GDF-15, troponin, VEGF, IGF1, CRP, IL6, TNF-alpha, and markers of the NO pathway. Thus, exploring common biological pathways underlying both atherosclerosis and cognitive decline could improve understanding of VCID. It is less clear if atherosclerosis, whether systemic or intracranial, is found in association with AD-specific neuropathologic changes. Although autopsy studies have reported associations between circle of Willis atherosclerosis and AD,¹²¹ this association has not been found consistently in studies using biomarkers of AD neuropathology in living subjects, with no association between intracranial atherosclerosis and amyloid by PET in nondemented participants from U.S. communities.¹²²

Importantly, the heart is not a surrogate for the brain when studying the contributions of atherosclerosis in VCID; therefore, careful studies to explore the specific dose-, age-, and gene-specific interactions in the impact of atherosclerosis and vascular risk factors on the brain are needed.^{123,124} For instance, the APOE4 allele appears to potentiate the adverse impact of a variety of vascular risk factors on the brain.^{125,126}

3.5 | Diabetes

Type 2 diabetes (T2D) is a growing public health problem affecting hundreds of millions of people globally. Economic forces driving lifestyle changes in physical activity and diet are contributing to the global growth of diabetes and related disease burden. Diabetes affects both small and large blood vessels that can lead to multiple end organ damage, including the nervous system.¹²⁷ In humans and non-humans, T2D is associated with changes in BBB permeability^{128,129} that are thought to contribute to cognitive impairment.⁵² Secondary inflammation that occurs in diabetes is also thought to further contribute to nervous system damage. T2D is associated with lower cortical volumes on MRI,¹³⁰ lower cognitive functioning,¹³¹ cognitive decline, and ADRD,^{132,133} however, mixed findings have been reported and mechanisms for observed neurocognitive dysfunction remain unclear.¹³⁴ T2D may be a more salient risk factor for non-AD pathology than for AD pathology, at least based on evidence from *post mortem* studies.^{135,136} Common co-occurring factors such as obesity, diet, and physical activity also appear to be associated with neurocognitive dysfunction, which further complicates the relationship between T2D and dementia.⁸⁶ Future research efforts should clarify the independent and interacting risks and causes that lead to both T2D and dementia to explain, and hopefully leverage, the link between these devastating disorders.

3.6 | Traumatic brain injury and blood brain barrier

There is growing awareness that contact sport athletes and military veterans who experience repetitive head impacts (RHI), including concussions and the asymptomatic smaller hits, subconcussions, are at risk for long-term neurologic problems including the development of chronic traumatic encephalopathy (CTE). There is accumulating evidence that exposure to RHI from contact sports and military

activity produces injury to the microvasculature, BBB dysfunction, and neuroinflammation.¹³⁷⁻¹⁴¹ Post-traumatic microvascular changes are also associated with impaired oligocyte development and alterations in white matter microstructure.^{142,143} CTE is a progressive neurodegenerative disease associated with repetitive head trauma that can only be definitely diagnosed by *post mortem* neuropathological analysis of brain tissue.¹⁴⁴⁻¹⁴⁶ The pathognomonic lesion of CTE is the accumulation of hyperphosphorylated tau (pTau) aggregates as neurofibrillary tangles and disordered neurites around small blood vessels at the depths of the cortical sulci.^{145,146} As CTE pathology becomes more severe, the perivascular pTau lesions become more numerous, affect more widespread brain regions, and eventually provoke a wide-scale neurodegeneration of the brain.¹⁴⁵ A β deposition, as plaques and amyloid angiopathy, is also found in aging subjects with CTE and there is evidence that the regional distribution of plaques and amyloid angiopathy in CTE differs from that found in normal aging and AD.^{147,148} Microvascular alterations, BBB dysfunction and neuroinflammation are severe and persistent at all stages of CTE pathology¹⁴⁹⁻¹⁵¹ and promote the continued accumulation of pTau.¹⁵² CTE is also associated with impairment of the glymphatic clearance system,^{153,154} which further serves to promote pTau accumulation and continued neurodegeneration.

4 | GAPS IN KNOWLEDGE AND RESEARCH OPPORTUNITIES

4.1 | Animal models

Previous reviews have summarized existing VCID animal models and remaining needs for animal models that recapitulate the human VCID disease spectrum.^{155,57} In this section, we will focus on models that mimic chronic cerebral hypoperfusion (CCH) as an example of VCID model. The increasing evidence from autopsy studies indicating that AD is due to mixed vascular and degenerative pathologies substantiates the need for adequate animal models that dissect the underlying mechanisms and to evaluate novel intervention strategies. Several cardiovascular risk factors (eg, hypertension, HHcy, and diabetes mellitus [DM]) have been associated with AD development.^{156,157} All of these risk factors can lead to CCH, which has been repeatedly linked to AD pathogenesis. To establish causal and temporal relationships several *in vivo* studies in mice have been conducted, which do indeed show that hypertension, DM, and HHcy have a similar impact on AD-related pathology and cognitive performance, which may be associated with cerebrovascular dysfunction.¹⁵⁸⁻¹⁶⁰ It is noteworthy that the number of mouse studies reviewing these risk factors in AD transgenic mice is limited and causal data are still rare. Because cerebral hemodynamic changes were not assessed in these animals, no data are available to indicate whether the aggravating effect of the vascular risk factors studies might be mediated by CCH. There is quite good evidence from mouse studies that indicates a role for hypoxia, induced by reduced cerebral flow, in AD neuropathology and reduced cognitive function. It is thought that hypoxia particularly stimulates APP processing and

A β production in ECs and neighboring neurons, promoting CAA generation and vascular dysfunction.¹⁶¹

Cardiovascular risk factors and cerebral flow disturbances have an accelerating and/or aggravating effect on AD brain pathology and cognitive decline in combined mouse models of vascular disease and AD. Based on current literature, we conclude that studies so far have not yet succeeded in establishing whether CCH plays a causative or coincidental role in AD disease initiation and/or progression. Underlying molecular mechanisms have not been clarified nor have therapeutic interventions been successful.

4.2 | Engineered human tissues

How vascular risk factors affect dementia is not understood and difficult to model in animals, as mice differ from humans in many ways, including A β clearance rate, APOE isoforms, lipoprotein profile, and the immune system.¹⁶² It is nearly impossible to fully recapitulate the many factors that contribute to AD in mice, and our reliance on mouse models may account for a significant proportion of the failure rate in translating animal study results into successful clinical trials. Given that clinical trials targeting the central nervous system (CNS) inherently have exceptionally high failure rates, it is imperative that alternative translational platforms with greater predictive value be developed and implemented for drug development. Current innovations across the health and engineering sectors are now yielding new platforms based upon the field of tissue engineering, where individual cellular components are combined together in 3-dimensional (3D) systems to generate functional tissue. Several new platforms are now available, mostly focused on the NVU that is composed of EC, smooth muscle cells (SMCs), pericytes, astrocytes, and neurons. Recent advances using induced pluripotent stem cells (iPSC)-derived neurons to form "brain in a dish" 3D cultures promise to fill an important niche between humans and animal models toward deciphering regulatory pathways and identifying therapeutic avenues for neurologic and psychiatric diseases.^{163,164} The most advanced model of human CAA to date was created using scaffold-directed 3D bioengineered vessels consisting of primary human ECs and human SMCs, cultured either without primary human astrocytes to mimic leptomeningeal arteries or with primary human astrocytes to mimic penetrating arteries.^{165,166} The engineered vessel contains a tight BBB that separates "brain" from "blood" circulating through the vessel lumen. Importantly, A β originating from the "brain" side crosses the engineered BBB into the "blood" side and both the rate of transport and accumulation in the vessel wall can be quantified. Further, the model recapitulates the well-established genetic effects of APOE on A β transport and accumulation. Suitability of this model for use in drug development was validated by investigating the beneficial effects of high density lipoprotein (HDL), which showed that HDL isolated from young healthy donors attenuated A β -mediated EC activation and reduced CAA in the engineered vascular wall independent of APOE isoform.^{165,166} These results show the promise of engineered human tissues as an experimental platform to investigate VCID mechanisms.

4.3 | Imaging tools and analysis

It is well recognized that dementia is often a heterogeneous disorder with mixed brain pathologies responsible for most cases of dementia in community-dwelling older adults.¹⁶⁷⁻¹⁷⁰ Some vascular contribution to the cognitive decline is present in almost all patients when their brains are assessed with sensitive MRI imaging methods during life or at autopsy and with the same burden of neurodegenerative disease, the presence of concomitant vascular brain injury increases the probability of clinical symptoms.^{44,171} This could be an additive effect of the two pathologies or a synergistic effect of ischemia on neurodegeneration.^{101,172-174} Hispanic and African American populations have a high prevalence of obesity, diabetes, and hypertension that make them particularly vulnerable to atherosclerosis and to VCID. Hence, quantifying, localizing, characterizing, and assessing the functional significance of a wide and increasing range of vascular brain injury markers through MRI or other imaging tools could help explain individual variations observed in susceptibility or resilience to cognitive decline and dementia. For example, studying the specific brain lesions associated with conditions causing premature vascular injury (voxel based lesion symptom mapping) could help us understand the structural basis for VCID in older adults.¹⁷⁵

Nowadays, there are extensive options of MRI imaging techniques that can be used to characterize cerebral vasculature and flow or functional properties, providing a valuable opportunity to enhance our understanding of VCID. A significant proportion of existing imaging studies have linked cerebral small vessel diseases (cSVD; such as white matter hyperintensity and microbleeds) to cognitive impairment.¹⁷⁶ In addition to the well-known role of cSVD in VCID, obtaining a more global picture of cerebral vasculature by including the evaluation of medium-to-large artery health may offer further insight. In this regard, established techniques such as vessel wall imaging¹⁷⁷ and emerging novel techniques such as angiography-based arterial features and functions (such as vessel length, number of branches, tortuosity, and flow dynamics) quantification tools would be of particular interest.^{178,179} It is also worthy to point out that serial imaging using these powerful techniques, rather than a single point "snapshot," should be considered in future studies because it can demonstrate the dynamic temporal relationship between vascular health and cognitive impairment.

4.4 | Biomarkers for VCID

As the NIA developed research priorities for the AD RD (specified as frontotemporal, Lewy body, vascular, and mixed dementia), one consistent recommendation emerged across all dementia types: to identify and validate biomarkers.⁸⁷ The rationale is that biomarkers can bridge the gap between preclinical models and human disease, providing windows on individual pathophysiologic processes that ultimately lead to clinical impairment. Biomarkers are also key to early phase clinical trials of novel or existing treatment strategies, facilitating patient

selection and risk stratification and providing actionable information on an intervention's target engagement, mechanism of action, and potential safety and efficacy.

NIH's response for the VCID field was to create the MarkVCID consortium (<https://markvcid.partners.org/w>).¹⁸⁰ Comprised of seven participating sites, a coordinating center, external advisory committee, and NIH leadership, MarkVCID's charge is to validate a suite of biomarkers for small vessel related VCID to the point of readiness for multi-center clinical trials. This mission has dictated MarkVCID's overall approach of focusing on biomarkers that are reproducible across sites and linked closely enough to key biological processes underlying cSVD to be useful for patient selection, risk stratification, or outcome assessment in a trial. An important corollary is that different VCID stages may require different biomarkers, such as markers of small vessel dysfunction in at-risk asymptomatic patients versus markers of white matter injury in symptomatic disease.

MarkVCID recently concluded its initial 2-year phase of harmonizing methods for clinical and cognitive assessment, neuroimaging, and fluid handling across the sites and implementing consent procedures to allow open sharing of samples and data. Each site has proposed candidate biomarkers that were reviewed by the coordinating center and external advisory committee and ultimately evaluated by NIH leadership for readiness and scientific priority. The remainder of MarkVCID's efforts will focus on independently validating the prioritized biomarkers (Table 3) across multiple consortium sites; thus, firmly establishing their availability for use in future interventional trials.

4.5 | Pathology markers for VCID

Further human pathologic studies are vital to advance knowledge of VCID. The mechanism of cognitive impairment from many vascular pathologies remains unclear and there is currently no animal model that mirrors the human condition. It is important to note that most neuropathology sampling schemes in aging studies focus on regions involved in neurodegenerative disease, yet vascular pathologies are often microscopic and multifocal so the presence as well as extent of vascular pathology is likely under-recognized in aging literature. For instance, microscopy of the aged human brain often shows a limited number of microinfarcts. Yet, just a few microinfarcts found at autopsy may indicate hundreds if not thousands in the whole brain,¹⁸¹ but even then it remains unclear if this amount of tissue loss fully explains cognitive impairment from microinfarcts.^{182,183} Indeed, in the human brain, there are both direct and indirect effects from vessel pathology and vascular tissue injury.^{184,185} Cellular and molecular mechanisms downstream of these vessel diseases and associated with tissue injury should be explored.¹⁸⁶ These downstream processes may include variations in inflammatory processes, clearance of waste, and interactions with other neurodegenerative processes. Vascular mechanisms may be focused on white matter¹⁸⁷ and/or the cortical regions of the brain,¹⁸⁸ and regional mechanisms could also differ. Indeed, watershed regions of the human brain appear especially important.¹⁸⁹ Linking vascular

TABLE 3 MarkVCID candidate biomarker kits^a

Imaging based biomarker kit	Lead research site
MRI peak skeletonized mean diffusivity	CHARGE Consortium
MRI white matter hyperintensity volume	Universities of California San Francisco, Davis, Los Angeles
MRI Arteriolosclerosis Score	Rush University, Illinois Institute of Technology
MRI white matter hyperintensity growth/regression	University of Kentucky
MRI free water	University of New Mexico Health Sciences Center, University of California Davis
MRI cerebrovascular reactivity	Johns Hopkins University School of Medicine
Optical coherence tomography angiography retinal vessel skeleton density	University of Southern California
Fluid based biomarker kit	Lead Research Site
Plasma endothelial signaling	Universities of California San Francisco, Davis, Los Angeles
Plasma exosome endothelial inflammation	Universities of California San Francisco, Davis, Los Angeles
Plasma neurofilament light chain	CHARGE Consortium
Cerebrospinal fluid placental growth factor	University of Kentucky

Abbreviation: VCID, vascular contributions to cognitive impairment and dementia; MRI, magnetic resonance imaging.

^aBiomarker kit protocols are available at <https://markvcid.partners.org/consortium-protocols-resources> (registration required).

pathologies to regional tissue injury, neurodegeneration, and cellular and molecular mechanisms in the human brain will be vital to advances in the field.

These vascular pathologies are often not visible or clinically recognized during life. While imaging can identify some of these pathologies, biomarkers for the human pathology have been elusive. For this reason, further well detailed prospective clinical-pathologic studies will be needed to sort out risk factors in late life for VCID until better biomarkers become available. Elevated and declining blood pressures in late life are associated with microinfarcts.^{184,190} Specific genetic factors and age may also play a role.¹⁹¹ Biomarkers for these intracranial vascular pathologies are critical to advance these epidemiologic studies.

Pathologic studies have shown that most older persons with cognitive impairment have multiple types of vascular disease in their brain with or without neurodegenerative pathologies.^{192,193} Given that model systems do not mirror the complexity of these complex phenotypes, using human tissue with mixed pathologies will be needed to explore and confirm proposed molecular and cellular mechanisms that can explain the common co-occurrence of vascular risk factors, pathologies, and proteinopathies, and neurodegeneration. New immunohistochemical multiplexing tools that can simultaneously interrogate different cellular and molecular markers within an intact spatial context are rapidly expanding and will be important in future neuropathologic studies.¹⁹⁴

4.6 | Tools/instruments needed for epidemiological studies

Epidemiological studies and clinical trials of dementia have traditionally focused on cognitive testing and generalized cognitive phenotypes

with limited assessments of neuropathology and subclinical vascular biomarkers. VCID can affect any and multiple cognitive domains and is most commonly associated with impairment in executive function, speed of processing, and attention.¹⁹⁵⁻¹⁹⁷ Standard screening instruments for global cognitive performance show strong discrimination for VCID at mild and severe impairment relative to unimpaired individuals,¹⁹⁸ but more work is needed. Development of cognitive batteries sensitive and specific to VCID declines may be needed to untangle VCID and mixed pathologies from forms of dementia without VCID. Longitudinal cognitive function remains the most clinically meaningful endpoint in research studies; yet, there are inherent limitations to relying solely on cognition. Cognition is proposed to be among the final pathophysiological changes in dementia.⁴⁷ Cognitive performance is inherently inferential with expectation that affected cognitive domains reflect pathophysiology of dementia; however, recent neuroimaging studies demonstrate that the clinical diagnosis of AD is biomarker negative in nearly 30% of cases.^{47,199} Given the long latency of dementia pathogenesis, a sizable portion of older adults will be biomarker positive and cognitively normal.²⁰⁰ Such findings resulted in a biomarker-based framework for research studies of AD.²⁰¹ Given evidence that cSVD and AD pathology are age-related phenomena that co-occur in most cases of dementia, it is essential that epidemiologic studies include comprehensive multimodal biomarker assessments to appreciate VCID.

Progress has been made to detect ADRD neuropathology using neuroimaging. Neuroimaging detection standards of cSVD on MRI are well established.²⁰² Continued MRI method development for quantifying microstructural integrity, functional changes (eg, brain networks, CBF, cerebrovascular reactivity), and automated cSVD detection are expected to further improve MRI assessments. Advances in PET imaging with ligands developed for fibrillar A β and neurofibrillary tau enable

TABLE 4 Subclinical vascular biomarkers and neuroimaging tools to better characterize VCID

Biomarker assessment level	Examples in VCID and ADRD studies		
	VCID—Peripheral	VCID—Brain	ADRD
Clinical events (passive < active surveillance)	Hypertension, CVD	Stroke, TIA	Dementia
Subclinical biomarkers (clinical, ambulatory)	Pulse wave, ABPM	Infarction, cSVD	Cognitive decline/MCI
Pathology	Atherosclerosis	Infarction	A β and tau
Functional measures	Endothelial function	CBF, distensibility	CSF clearance

Abbreviations: ABPM, ambulatory blood pressure monitoring; ADRD, Alzheimer's disease-related dementias; CBF, cerebral blood flow; CSF, cerebral spinal fluid; cSVD, cerebral small vessel diseases; CVD, cardiovascular disease; MCI, mild cognitive impairment; TIA, transient ischemic attack; VCID, vascular contributions to cognitive impairment and dementia.

comprehensive assessment of AD pathology and differentiation from cSVD in multimodal neuroimaging studies. As such, vascular risk factors (eg, midlife risk scores, BP,²⁰³⁻²⁰⁷ genetic variants, and arterial stiffness^{88,208,209}) have been associated with greater A β deposition, A β progression over time, and tau burden. Two caveats deserve consideration. First, PET imaging ligands demonstrate non-specific binding in the white matter, although reduced binding is being evaluated as a potential vascular biomarker.²¹⁰ Second, currently available PET ligands designed to bind fibrillar A β cross the BBB and cannot differentiate vascular and parenchymal A β ;²¹¹ CAA-specific PET tracers hold promise for differentiating vascular and parenchymal deposition of A β .²¹² Subclinical vascular biomarkers and neuroimaging tools can be used to better characterize the pathology connecting vascular factors to dementia (Table 4).

4.7 | Evaluation of life-course exposure and its impact

Epidemiologic studies have contributed key insights into the association between cardiovascular risk factors and risk of cognitive impairment and dementia over the life course. Such studies have consistently demonstrated that mid-life cardiovascular factors,^{17,34,213,214} including diabetes,²¹⁵⁻²¹⁷ blood pressure/hypertension,^{31,32,36,218} and obesity^{33,123,219-221} increase risk of late-life cognitive impairment and dementia. In contrast, results for late-life cardiovascular factors have been less clear with some studies finding weaker associations or in the case of high blood pressure and obesity, converse associations in which late-life cardiovascular factors were associated with lower risk of adverse cognitive outcomes.^{123,215,219-221} The true role of late-life risk factors on cognition can be hard to evaluate, given concerns about reverse causation (in which changes in blood pressure and weight, for example, may be secondary to cognitive impairment, for example, or at least indicate concurrent disability which could contribute to risk of dementia),^{221,222} rather than themselves causing cognitive decline. Furthermore, survival bias reduces the likelihood that individuals with markedly abnormal vascular risk factors in mid life survive to late life, both limiting their ability to develop dementia,²²³ and reducing the likelihood of these risk factors remaining in the very elevated range in older adulthood.

Evidence is now emerging that establishes the importance of cardiovascular risk factors for cognitive outcomes even earlier in the life course. In the Coronary Artery Risk Development in Young Adults Study, cardiovascular factors beginning from early adulthood,¹²⁰ including systolic and diastolic blood pressure, glucose, subclinical atherosclerotic calcification, and heart rate variability, were associated with cognitive outcomes at mid life.^{120,224-229} In the Young Finns Study, childhood cardiovascular factors have also been linked to mid-life cognitive function even after adjusting for mid-life cardiovascular exposures.²³⁰ Assessment of vascular risk factors over time also allows for evaluation of a range of cardiovascular risk factor characteristics including clinical as well as subclinical risk factors levels,^{120,224} cumulative exposure,^{224,230} trajectories, and changes or variability over time.^{226,228}

Critical questions remain regarding the role of vascular risk factors over the life course, particularly the mechanistic pathways by which cardiovascular risk factors influence cognitive outcomes at each life stage, the link between mid-life cognitive changes and the development of late-life vascular cognitive impairment and dementia, and the effects of treatment timing on prevention/delay of cognitive impairment. In addition to traditional observational studies, researchers are also leveraging the power of existing epidemiological studies and combining data across cohorts and life course stages to address these crucial questions.

4.8 | Assessment of environmental and social factors

In the accompanying sections, we have described the cerebrovascular and cardiovascular disease (CVD) risks, biomarkers, and pathologic markers of VCID. In addition, it is important to recognize that our environment, communities, and socioeconomic conditions impact our vascular, neurologic health, and disease.¹⁶⁸ The known environmental and sociocultural factors associated with CVD and stroke can also be assessed as contributors to VCID. Arguably, the constant and high demands for blood supply of brain tissue makes it exquisitely sensitive to acute and chronic blood flow disruptions that are known to affect cognitive function and ADRD.²³¹ Environmental factors, such as air pollution, are recognized risks shared by CVD and dementia.²³²

Socioeconomic level and ethnicity/race are related to where you live, and the healthiness of your community. Limited access to healthy and safe environments for socioeconomically disadvantaged communities can hinder physical activity, which is essential to maintain a healthy vascular system that is vital to brain health. Access to affordable, high-quality foods that are important for lifespan brain development and maintenance also vary significantly by a community's socioeconomic level and ethnicity/race. Environmental and social factors should be included in the study of VCID because they can inform public health initiatives for reducing ADRD burden and brain health.²³³

NHLBI and NINDS have long supported longitudinal cohort studies for examining CVD in relation to environmental and social factors in diverse populations, and the National Institute on Aging has efficiently leveraged these cohorts to understand the aging brain and disorders. As such, these cohort studies are rich resources replete with sociocultural information and deep phenotyping and genotyping of vascular health and disease, and VCID. These longitudinal data are accessible and facilitate lifespan studies of when and how vascular health and disease impact cognitive health and diseases. In terms of public health, these available data resources can guide study of optimal adult lifespan time points for developing targeted disease prevention and therapeutic interventions for reducing VCID.

4.9 | Future clinical trials

It is increasingly clear that ADRD, like many biologic processes, is not the result of a solitary factor but instead due to a complex, multi-step combination of physiologic events. Incident ADRD is associated with vascular, neurologic, metabolic, and inflammatory signals and responses.^{44,234} However, clinical trials to date have typically tested interventions based on the basic science observations focused on the maintenance of neuronal cell function and prevention of neuronal death as a pathway for reducing ADRD burden. More recently, the emergence of physiologic evidence supporting the diverse pathways involved in the process of ADRD onset has broadened the opportunity for developing clinical trials focused on additional targets, including vascular targets.²³⁵ In addition, the recent imaging results from clinical trials such as ACCORD (Action to Control Cardiovascular Risk in Diabetes) and SPRINT-MIND indicate that interventions which reduce CVD risk also reduce the rate of abnormal white matter disease formation, providing imaging evidence for the biologic basis of an intervention and an effective surrogate outcome for efficacy.^{40,236} Although no significant difference in the primary outcome, dementia, was found with more aggressive blood pressure control, SPRINT-MIND made an additional important advance in the field by showing the feasibility of including adjudicated MCI as a clinical trial outcome and the need for composite outcomes (MCI and dementia) when event rates are low, thus improving the power and sample size metrics needed to test future interventions. These and other results highlight the need for additional investment in understanding the basic mechanism for the efficacy of cardiovascular disease prevention therapies, such as improved blood pressure control, on lowering the risk for MCI

and ADRD. Animal models from rodents to non-human primates are critical to inform the epidemiologic and clinical trial observations in humans as a foundation for the design of additional definitive clinical trials focused on impacting vascular mechanisms in ADRD.

4.10 | Limitations of the convened workgroup and this report

This report summarized the topics that were discussed at the workshop and is not a comprehensive review of the state of the field. We acknowledge that several important topics and contributors to VCID were not discussed at the workshop and thus not included in this report, but were included in the 2019 ADRD summit report. These topics include, but are not limited to, the following: the role of psychosocial factors, including stress, depression, and mental health; sleep and sleep disorders; and importantly, social determinants of health and their impact on cognitive outcomes as they relate to VCID. In addition, several forms of VCID and possible mechanisms were not discussed or were only discussed minimally at the WG, and thus not included in this report. These include (among others): cerebral amyloid angiopathy as a form of VCID, its subtypes, and its overlap with AD; the role of perivascular and glymphatic drainage in VCID development or progression; and overlapping pathology with cardiac disease including the previously described finding of cardiomyocyte inclusions in some dementing illnesses. Some of these topics represent important research priorities and opportunities; we discuss below those opportunities that were specifically discussed at the WG, but acknowledge the large number of additional relevant areas important for future study.

5 | RESEARCH OPPORTUNITIES

The following research opportunities were identified on the basis of the presentations given at the workshop, together with discussion among attendees.

Basic mechanisms and experimental models

- Understand the contribution of endothelial and pericyte dysfunction to dementia
- Understand the contribution of BBB breakdown to dementia
- Comprehensively assess vascular pathologies in the brain and elsewhere that are associated with dementia
- Develop and validate clinically relevant blood-based biomarkers for vascular dementia that can offer opportunities for early detection and risk stratification, as well as monitoring disease progression and individual response to intervention
- Understand the effects of systemic inflammation on cognitive impairment and dementia, develop tools to measure systemic inflammation, and understand the potential therapeutic value of immunotherapy for VCID
- Develop and validate mammalian animal models of VCID that recapitulate human disease progression and brain pathologies

- Develop and validate next-generation human-based in vitro models to investigate VCID mechanisms

Human-based studies

- Develop and validate non-invasive imaging techniques to monitor subtle cerebrovascular and blood flow deficits, BBB permeability, and other subclinical pathologies that can be used as surrogate markers for dementia in humans
- Develop and validate neuropsychological batteries that are helpful in the context of VCID whether it occurs alone or in combination with other dementia risk factors for use in epidemiological studies of VCID and clinical trials
- Develop and validate new molecular biomarkers of blood vessel injury, including pericytes and endothelial cell injury and BBB breakdown, and biomarkers for neurovascular dysfunction associated with human cognitive impairment
- Understand the effects of the timing of onset and duration of vascular risk factors on cognitive impairment and dementia
- Develop and validate risk stratification strategies during optimal time points over the life course (eg, mid life) in longitudinal cohort studies
- Understand the variations in dementia risk at the population and individual level: biological, psychological, and socioeconomic factors that account for differences in incidence and trajectory of cognitive function decline in different racial and ethnic groups, and sex differences
- Understand the interrelationships among aging, sex, race, ethnicity, and social determinants of health (economic stability, education, social and community context, health and health care, and neighborhood and built environment), and how these factors contribute to the development, progression, and management of VCID
- Add CVD risk measurements in dementia cohort studies and add cognitive function, brain imaging (MRI, functional MRI, PET), mood, and behavioral function measurements in cardiovascular cohorts/trials when feasible
- Understand the influence of sleep on vascular dysfunctions and cognitive function, or the effect of sleep disorders on cognitive impairment and dementia
- Understand the impact of atrial fibrillation and heart failure on cognitive impairment and dementia

Develop methodologies, workforce, and shared resources

- Support the development and validation of novel imaging techniques in animals and humans that permit monitoring of cerebral and peripheral vascular function and detecting vascular changes prior to the onset of dementia
- Support the establishment of standard clinical and pathological protocols to measure vascular changes and cognitive function that can be applied to diverse populations both within the United States, across different socioeconomic statuses with different access to health care, and across countries

- Facilitate cross-cohort, cross-institutional, and cross-discipline collaboration to consolidate effort and merge available data on VCID
- Support training and workforce development and encourage collaboration in neurology, cardiovascular research, social and behavioral science, clinical trials, and implementation research
- Develop and validate simple clinical tools to access and grade MRI findings consistent with VCID

Some of these research gaps may need to be addressed at a global level and through public-private partnerships, as proposed at the 2018 World Health Summit.²³⁷ In summary, collaboration and cross-talk across institutes and disciplines; exploring common biology, as well as differences underlying cardio- and cerebrovascular diseases and VCID; harmonizing methods and leveraging existing cohort studies and clinical trials to include cognitive end points in cardio- and cerebrovascular prevention studies and detailed vascular risk factor information in dementia prevention studies are all recommendations to hasten our understanding and develop more effective prevention and treatment of VCID.

FUNDING INFORMATION

- This workshop was supported by the National Heart, Lung, and Blood Institute (NHLBI) and National Institute of Neurological Disorders and Stroke (NINDS), and National Institutes of Health (NIH).
- The work of B.V.Z. is supported by the NIH grants R01AG023084, R01NS090904, R01NS034467, R01AG039452, 1R01NS100459, 5P01AG052350, and 5P50AG005142 in addition to the Alzheimer's Association VCID-17-509279, Cure Alzheimer's Fund, and the Foundation Leducq Transatlantic Network of Excellence for the Study of Perivascular Spaces in Small Vessel Disease reference no. 16 CVD 05.
- The work of Z.S.K. is supported by the NIH grants R01HL111062, R01HL131515 and by the Mayo Foundation.
- The work of T.M.H. is supported by the NIH grants R01AG054069, R01AG053938, R01AG054491, R01AG018915-16S1, R01AG058969, RF1NS110043, R03AG064569, U01HL096814, and P30AG049638.
- The work of S.S. is supported by the NIH grants R01 NS017950 and R01 AG054076.
- The work of K.Y. is supported by NIH grants R01AG063887 and RF1AG054443.
- The work of H.M.G. is supported by the NIH grants R01AG048642, RF1054548, RF10621022, P30AG062429, and P30AG059299.

ACKNOWLEDGMENT

We acknowledge Servier Medical Art (<https://smart.servier.com>) for providing images for **Figure 1**.

CONFLICTS OF INTEREST

T.M.H. has no conflicts to report. R.F.G. was previously an Associate Editor for the journal *Neurology*. H.S. is a full-time employee of the Alzheimer's Association. H.M.G. is an Associate Editor for the journal *Alzheimer's & Dementia*. Any opinions, findings, conclusions, or

recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Heart, Lung, and Blood Institute; National Institute of Neurological Disorders and Stroke; National Institute on Aging; or the National Institutes of Health.

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How to cite this article: Zlokovic BV, Gottesman RF, Bernstein KE, et al. Vascular contributions to cognitive impairment and dementia (VCID): A report from the 2018 National Heart, Lung, and Blood Institute and National Institute of Neurological Disorders and Stroke Workshop. *Alzheimer's Dement*. 2020;16:1714-1733. <https://doi.org/10.1002/alz.12157>